

Thanh Bao TRAN

M2 Master Student, Grenoble INP, UGA, Grenoble, France
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EDUCATION

Master of Science in Mobile, Autonomous and Robotic Systems

Grenoble INP — University of Grenoble Alpes, Grenoble, France

GPA: 14.62/20.00

09/2025 — 07/2026

Relevant Coursework: Learning MPC (Optimal Control, MPC, Reinforcement Learning), Mobile Robotics (Perception, UAV Control, State Estimation), Autonomous Systems (Path Planning, Multi-Robot Cooperation).

Degree of Engineer in Control Engineering and Automation

Hanoi University of Science and Technology, Hanoi, Vietnam

GPA: 3.05/4.00

08/2018 — 09/2023

Relevant Coursework: Control Theory, Robotics, Nonlinear Systems, Embedded Systems.

Exchange Program in Electrical Engineering and Information Technology

Technical University of Munich, Munich, Germany

Erasmus+ Scholar

10/2021 — 04/2022

Relevant Coursework: Advanced Robot Control and Learning, Power Electronics.

PUBLICATIONS

- **Tran Thanh Bao**, Nguyen Minh Duc, Minh Duc Duong, and Trong-Hieu Do, “ZV Shaper – ADRC Combination Control for Crane System with Constrained Control Signal”, *Mathematical Modeling and Computing in Automation*, vol. 4, no. 3, pp. 32–38, Dec. 2023.

RESEARCH EXPERIENCE

Research Intern

DANCE Team, Inria Grenoble (Advisor: Prof. Paolo FRASCA)

Grenoble, France

02/2026 — 07/2026

- Researching control of connected & autonomous vehicles (CAVs) for stochastic traffic flow systems, with focus on scalable density-feedback controllers, asymmetric stochastic car-following models, and string-stability analysis under driver heterogeneity.
- Owning end-to-end research pipeline: stochastic microscopic simulator with 2 HV models and 3 AV controllers; analytical stability proofs; empirical sweep validation; literature cross-verification.

Robotics Research Engineer

Apicoo Robotics

Hanoi, Vietnam

04/2024 — 09/2024

- Simulated robotic arm controls in Isaac Sim for Sim2Real applications.
- Developed a 6D object-pose pipeline for grasping: RGB-D acquisition via the Intel RealSense SDK, DBSCAN + ICP (Open3D/PCL) as a classical baseline, then learning-based estimators (DOPE for 4 packaging-type objects — carton box, beverage can, plastic bottle, jar; FoundationPose/MegaPose evaluated for novel objects).
- Achieved ~85% grasp success over ~100 trials at ~20s per pick cycle on a real arm with an eye-in-hand RealSense camera, solving hand-eye calibration.

Research Assistant

CIR Lab, National Taiwan Normal University

Taipei, Taiwan

10/2023 — 04/2024

- Developed a dual-arm bimanual manipulation system (bottle-cap opening) using deep reinforcement learning (PPO) on top of Operational Space Control in NVIDIA Isaac Gym.
- Trained a custom pose CNN (4-channel RGB-D ResNet18) on ~10k domain-randomized images rendered via the Isaac Gym camera API, localizing the bottle and cap to a median error of ~2.2 mm, feeding the policy observation.
- Designed and tuned the observation/action spaces and reward function for the PPO policy commanding 6-DoF delta-poses per arm.
- Closed the full perception-to-policy-to-control loop in simulation and initiated transfer to a real dual Franka Emika setup with a RealSense camera, tackling hand-eye calibration.

Undergraduate Research Assistant

RA Lab, Hanoi University of Science and Technology

Hanoi, Vietnam

05/2021 — 09/2023

- Designed and implemented Active Disturbance Rejection Control (ADRC) combined with zero-vibration input-shaping for overhead crane systems.
- Conducted both simulation studies and hardware experiments with real-time embedded controllers.

RESEARCH PROJECTS

Microscopic and macroscopic models for the control of mixed-autonomy traffic
Internship Research Project (Advisor: Prof. Paolo FRASCA)

Grenoble, France
02/2025 — 07/2026

- Reproduced Stern et al. 2018 ring-road experiments (FollowerStopper, PI Controller).
- Designed and tuned a density-feedback “Cloud P” controller with a safety filter; via grid-search derived scaling law $K_{\text{look}} = \lceil N/5 \rceil$, achieving 100% σ_v reduction over PI baseline across $N=20\text{--}200$ vehicles.
- Performed string-stability analysis of the Stochastic Asymmetric OV-FTL model (asymmetric coupling + Ornstein–Uhlenbeck noise) via block-circulant Fourier diagonalization & Maximum Modulus Principle.

Flatness-Based Trajectory Planning and Control for UAV-Slung Load Systems
Advanced Research Project

Grenoble, France
09/2025 — 02/2026

- Designed a cascade control architecture for a quadrotor slung-load system: flatness-based trajectory planning, position ADRC with a known-unknown decomposition of the cable force (nominal tension as feedforward, ESO rejecting only deviations), and geometric attitude control on SO(3).
- Implemented a 10-state Extended Kalman Filter fusing position, cable-angle, and IMU measurements to estimate the full state and cable tension; IMU fusion reduced velocity-estimation RMSE by $\sim 44\%$ over position-only sensing.
- Validated in MATLAB simulation under realistic sensor noise across point-to-point, circular, and figure-8 trajectories.

Following Me: LiDAR-Based Person Following on a Mobile Robot
Mobile Robotics Course Project

Grenoble, France
09/2025 — 02/2026

- Built a real-time (10 Hz) detection–tracking–control pipeline in ROS2 (C++) on the RobAIR platform (Hokuyo UST-15LX 2D LiDAR): background-subtraction motion detection, scan clustering, and leg-pair person detection.
- Implemented a nearest-neighbor data-association tracker with an adaptive uncertainty radius and a track-persistence counter, robust to occlusion and sensor noise.
- Designed dual PID control (rotation + translation) with velocity limits (max 0.6 m/s), a 0.35 m safety distance, and closest-obstacle avoidance; commanded the robot over Wi-Fi via distributed ROS2 (DDS).
- Validated on recorded rosbags (9 detection / 5 tracking) and 3 scripted failure scenarios; analyzed the target-switching limitation of greedy nearest-neighbor association and fixed a phantom-person detection bug via a leg-exclusivity constraint.

Learning Model Predictive Control for Autonomous Racing
L-MPC Coursework Project (Advisor: Prof. Luigi GLIELMO)

Grenoble, France
09/2025 — 02/2026

- Implemented Learning-MPC framework for autonomous racing.
- Combined model predictive control with iterative learning with safe set-based value function approximation for optimal lap time performance.

HONORS AND AWARDS

Academic Excellence Scholarship

Awarded by Hanoi University of Science and Technology for outstanding academic performance

Hanoi, Vietnam
02/2023

TUMexchange Scholarship (Erasmus+)

Highly competitive EU scholarship for academic excellence (1 of 4 students selected in Vietnam)

München, Germany
07/2021

TECHNICAL SKILLS

- **Programming:** Python, C++, MATLAB/Simulink
- **Robotics Software:** ROS2, NVIDIA Isaac Gym/Sim, Gazebo, RViz, Autoware (basic)
- **ML & CV:** PyTorch, OpenCV, Open3D/PCL, scikit-learn, DRL-PPO, 6D pose estimation (DOPE, MegaPose, FoundationPose)
- **Control & Optimization:** MPC, LQR, ADRC, Optimal Control, Convex Optimization, EKF
- **Hardware Integration:** UAV flight controllers, IMU sensors, depth cameras (Intel RealSense SDK / pyrealsense2)
- **Development Tools:** Git, Linux, Docker, Blender

REFERENCES

Prof. Paolo FRASCA

CNRS Researcher (HDR), GIPSA-lab & Inria, Grenoble, France
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Prof. Luigi GLIELMO

Professor of Automatic Control, University of Naples Federico II, Italy
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Prof. Trong-Hieu DO

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